

Computation of Erdős-Woods Numbers.

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Erdős-Woods Numbers are defined as the length of an interval of consecutive integers whose every element is not coprime with its extremities. Woods was the first to find such numbers, Dowe proved there exists an infinity and Cégielski, Herout and Richard that their set is recursive. Our aim is to study the arithmetical proprieties of those numbers.

History and Conjecture

Conjecture Erdős-Woods : There is a positive integer k such that every x is uniquely determined by the list of prime divisors of $x + 1, x + 2, \dots, x + k$.

Perhaps $k = 3$.

For primes less than 23 there are four ambiguous cases for $k = 2$:
 $(x + 1, x + 2) = (2, 3)$ ou $(8, 9)$; $(14, 15)$ ou $(224, 225)$; $(75, 76)$ or $(1215, 1216)$.

The first three of these are members of the infinite family :

$$(2^n - 2, 2^n - 1), (2^n(2^n - 2), (2^n - 1)^2)$$

Conjecture, Counterexample and Theorem

Conjecture (Woods 1981) : For any ordered pair $\langle a, d \rangle$ of natural numbers, with $d \geq 3$, there exists a natural number c such that $a < c < a + d$ and c is coprime with a and with $a + d$. In other words:

$$\forall a, \forall d > 2, \exists c [a < c < a + d \wedge a \perp c \wedge c \perp a + d].$$

2184

Counterexample (Woods) : $\langle 2184, 16 \rangle$

2184	2185	2186	2187	2188	2189	2190	2191
2,3,7,13	5,19,23	2,1093	3	2,547	11,199	2,3,5,73	7,313
2192	2193	2194	215	2196	2197	2198	2199
2,137	3,17,43	2,1097	5,439	2,3,61	13	2,7,157	3,733
2200							
2,5,11							

Theorem (Dowe 1989) : There exists an infinity of such counterexamples to the Woods'conjecture.

Elementary Propositions

Proposition 1 : If d is an Erdős-Woods number there exists two primes dividing $d - 1$.

Idea : Primality of consecutive primes.

Proposition 2 : a and $a + d$ are composite numbers.

Idea : Primality of consecutive primes.

Remark : d can be a prime.

Elementary Propositions

Proposition 3 : The complementary set in $\mathbf{N}$ of the set of Erdős-Woods number is infinite.

Idea : For every prime p , $p^n + 1$ belongs to this set.

Proposition 4 : Pour tout couple d'entiers a et d , tels que d soit un nombre de Erdős-Woods on a $d < a$.

Idea : Bertrand-Tchebychev's Theorem

Theorem (Cégielski-Heroult-Richard 2000)

An integer d is an Erdős-Woods number if and only if there exist a partition of the set $P_{<d}$ of primes (strictly) less than d in two sets A and B , and an application P from $[1, d - 1]$ on $P_{<d}$ such that:

- (i) for any integer i , $1 \leq i < d$, if $P(i) \in A$ then $P(i) \mid i$ and $P(i) \mid a$, if $P(i) \in B$ then $P(i) \mid d - 1$ and $P(i) \mid a + d$;
- (ii) for $1 \leq i < i + P(i) < d$, we have $P(i) \in B$ iff $p(i + P(i)) \in B$

Idea : Analogous to the Cantor Theorem on consecutive primes in arithmetic progressions.

First Erdős-Woods numbers

16, 22, 34, 36, 46, 56, 64, 66, 70, 76, 78, 86, 88, 92, 94, 96,
100, 106, 112, 116, 118, 120, 124, 130, 134, 142, 144, 146,
154, 160, 162, 186, 190, 196, 204, 210, 216, 218, 220, 222,
232, 238, 246, 248, 250, 256, 260, 262, 268, 276, 280, 286,
288, 292, 296, 298, 300, 302, 306, 310, 316, 320, 324, 326,
328, 330, 336, 340, 342, 346, 356, 366, 372, 378, 382, 394,
396, 400, 404, 406, 408, 414, 416, 424, 426, 428, 430, 438,
446, 454, 456, 466, 470, 472, 474, 476, 484, 486, 490, 494,
498, 512, 516, 518, 520, 526, 528, 532, 534, 536, 538, 540.

Open Problems

Let $EW(n)$ be the n -th Erdős-Woods number. $EW(1) = 16$, $EW(2) = 22$, $EW(3) = 34$, ...

Problem : Find a lower bound for $EW(n)$.

Problem : Find a (simple) function f such that $EW(n) \sim f(n)$.

Problem : Is the density $\rho(n)$ of Erdős-Woods numbers linear? More precisely: $\rho(n) = O(n)$?

Perhaps $\frac{\rho(n)}{n} \rightarrow \frac{1}{4}$

Conjectures

Conjecture (Dowe 1989) : Every Erdős-Woods number is even.

Conjecture (Dowe 1989) : Every a associated to Erdős-Woods numbers is even.

Conjecture (CHR 2000) : There exists an infinity of twins Erdős-Woods numbers.

Conjectures

Conjecture (CHR 2000) : For any integer k , there exists an integer d such that $d, d + 2, d + 4, \dots, d + 2.k$ are Erdős-Woods numbers.

Conjecture (Vsemirnov 2000) : Every square greater than 16 is an Erdős-Woods number.

Results

Theorem : If for an odd integer number d , there exists a sequence : $S_d = d_0, d_1, d_2, \dots$ such that:

$$* d_0 = 1$$

$$* \forall 1 \leq i \leq k, d_i \text{ is a prime factor of } d - d_{i-1}$$

$$* \forall 1 \leq i \leq k, d_i^2 > d$$

$$* \exists i \leq k, d_k = d_i$$

then d is an Erdős-Woods number.

Corollary : 903, 2545, 4533, ... are Erdős-Woods numbers.

Ideas

903. 41, 431, 59, 211, 173, 73, 83.

2545. 53, 89, 307, 373, 181, 197, 587.

4533. 103, 443, 409, 1031.

903, [[41, 39], [431, 22], [59, 43], [211, 84], [173, 139], [73, 58],
[83, 27]]

2545, [[53, 13], [89, 4], [307, 135], [373, 207], [181, 45], [197, 67],
[587, 100]]

4533, [[103, 48], [443, 331], [409, 252], [1031, 92]]

New Ideas

Theorem : If for an even integer number d^2 with $d+1$ prime, there exists a sequence : $S_{d^2} = d_0, d_1, d_2, \dots$ such that :

$$* d_0 = 1$$

$$* d_1 = d + 1$$

$$* \forall 2 \leq i \leq k, d_i \text{ is a prime factor of } d^2 - d_{i-1}$$

$$* \forall 2 \leq i \leq k, d_i^2 > d^2$$

$$* \exists i \leq k, d_k = d_i$$

then d^2 is an Erdős-Woods number.

Remark : Vsemirnov's conjecture is false for 26^2 and 34^2 .

Corollaries and Certificates

Corollary : 16, 36, 100, 144, 256, 324, 484, 784, 900, 1296, 1600, 1764, 2116, 2704,... are Erdős-Woods numbers.

Certificates 16, [[5, 1], [11, 6]]

36, [[7, 1], [29, 6]]

100, [[11, 4], [89, 30]]

144, [[13, 4], [131, 75]]

256, [[17, 1], [239, 221]]

324, [[19, 10], [61, 31], [263, 60]]

Certificates

484, [[23, 4], [461, 206]]

784, [[29, 14], [151, 40], [211, 16], [191, 41], [593, 15]]

900, [[31, 3], [79, 52], [821, 364]]

1296, [[37, 17], [1259, 989]]

1600, [[41, 17], [1559, 1055]]

1764, [[43, 11], [1721, 403]]

2116, [[47, 4], [2069, 1068]]

2704, [[53, 3], [241, 138], [821, 667], [269, 230], [487, 192], [739, 510], [131, 68], [83, 29], [2621, 1506]]

Consecutive Erdős-Woods numbers

$n=2$: 34, 36

$n=3$: 92, 94, 96

$n=4$: 216, 218, 220, 222

$n=5$: 532, 534, 536, 538, 540

$n=6$: 1834, 1836, 1838, 1840, 1842, 1844

$n=7$: 2166, 2168, 2170, 2172, 2174, 2176, 2178

$n=8$: 4312, 4314, 4316, 4318, 4320, 4322, 4324, 4326

$n=9$: 4032, 4034, 4036, 4038, 4040, 4042, 4044, 4046, 4048